

SHIVA NUNEMUNTHALA

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Experience

Software Engineer

Jun 2023 – Present

Bluestone

Bengaluru

- Engineered a low-latency web security layer with rule-based detection, CIDR-based blocking, rate limiting, and heuristic request scoring to mitigate **XSS** and **SQLi** attacks, screening **10M+ daily requests** with under **3 ms** latency.
- Redesigned a monolithic application into an **Identity microservice** using **Spring Boot** and **Hibernate** with a dedicated **MySQL** database, leading the migration of authentication, authorization, and customer-specific access logic.
- Implemented a real-time observability and security monitoring pipeline with **Wazuh**, **Elasticsearch**, and **Kibana**, processing **50+ GB of logs daily** for automated alerting across infrastructure and application-level anomalies.
- Architected an AWS data pipeline using **Glue ETL** to sync **30+ production MySQL tables** into Apache Iceberg on S3, powering the AI analytics engine's queries via **Athena**.
- Built a product price-tracking system on **MongoDB**, capturing daily price history and customer view activity across the catalog, and triggering automated re-engagement alerts to customers on significant price drops since their last visit.
- Developed fully automated analytics pipelines aggregating product views, purchases, and high-value customer metrics into scheduled reports, powering the marketing team's campaign targeting without any manual reporting effort.

Project Associate

Sep 2025 – Feb 2026

Indian Institute of Technology Indore

— Research Advisor: Dr. Nagendra Kumar

Remote, Part-time

- Devised a dual-encoder semantic segmentation architecture with **ConvNeXtV2-based** and **Vision Mamba-based** branches to capture complementary local features and long-range dependencies.
- Enhanced the **SS2D block** using **fractal scanning strategies** for improved multi-scale context modeling, and proposed an **Attention-Guided Global-Local Feature Fusion module** for effective cross-branch feature integration.
- Designed a class-aware probabilistic instance-level **Copy-Paste Augmentation** strategy to mitigate class imbalance and improve minority-class learning, generating augmented samples with **FID < 8** across all evaluated datasets.
- Achieved **~4.5× reduction in GFLOPs** vs. benchmark models while maintaining competitive segmentation performance. **Co-first author on a manuscript under review.**

Machine Learning Research Intern

Aug 2022 – Nov 2022

Math of Data Science and Simulation Lab

— Research Advisor: Prof. Kapil Ahuja

Indore

- Constructed a thyroid nodule classification framework leveraging **Pass-Band Discrete Cosine Transform (PB-DCT)** based texture descriptors and addressed class imbalance with **SMOTE** to enhance classification efficiency.
- Integrated multiple **model-agnostic** explainability techniques such as **SHAP**, **LIME**, **CBR**, and **Anchors**, to enhance model transparency, interpretability, and clinical decision support.
- Achieved an accuracy of 99.7% while preserving strong **interpretability** and **generalization** across imbalanced data.

Software Engineer Intern

Jan 2022 – Jun 2022

Anastrat

Bengaluru

- Built tools to evaluate multi-leg **options strategies**, assessing payoff across strike and expiry combinations.
- Developed algorithms mapping broker-specific symbols to market symbols, and refactored core backend modules.

Education

Indian Institute of Technology Indore

Aug. 2019 – May 2023

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.3/10

Achievements

- Accomplished AIR **1048** in the **Joint Entrance Examination (JEE)** among a cohort of 1.3 million+ students.
- State rank 12 in **Regional Mathematics Olympiad** and qualified for **Indian National Mathematics Olympiad**.
- National top 1% in **NSEC** and qualified for **Indian National Chemistry Olympiad (INChO)**.
- All India Rank **1452** (top 0.7%) in **Kishore Vaigyanik Protsahan Yojana (KVPY)** fellowship test by IISc Bangalore.

Projects

Single Vehicle Cyclic Inventory Routing Problem | *Python, Convex Optimization* **Sep 2022 - Nov 2022**

- Reformulated the SV-CIRP using **continuous relaxation** to obtain a **convex optimization** formulation, and implemented Mixed-Integer Programming (MIP) solvers to compute feasible solutions.

Interpretability of Brain Cancer Segmentation | *Python, explainable AI* **Nov 2022 - Jan 2023**

- Incorporated **Layerwise Relevance Propagation** to provide insightful explanations for black-box model predictions.
- Utilized advanced **XAI** techniques, including **Gradient-weighted Class Activation Mapping** and **SmoothGrad**, to interpret and analyze segmentation models effectively.

Sparse Dynamic Programming for the 0/1 Knapsack Problem | *C++, Parallel Computing* **Aug 2021 - Dec 2021**

- Parallelized the Sparse Knapsack Dynamic Programming (SKPDP) algorithm using **fine** and **coarse-grained** parallelism, achieving computational complexity **independent of capacity size** for large, weakly correlated instances.

Speedup Matrix Multiplication and Decomposition | *C++, Parallel Computing* **Aug 2021 - Dec 2021**

- Utilized **OpenMP** to optimize **LU decomposition** of large matrices, harnessing **parallelism** to significantly boost computation speed.
- Analyzed and evaluated various communication methods of **MPI**, including **blocking**, **non-blocking**, and collective communication methods, to enhance matrix multiplication performance.

Technical Skills

Languages: Java, Python, C++, SQL

Backend: Spring Boot, Hibernate, MySQL, MongoDB, Elasticsearch

Data & Cloud: AWS (Glue, Athena, S3), Apache Iceberg, Spark, Wazuh, Kibana

ML: PyTorch, scikit-learn, NumPy, Pandas